

# INTERNSHIP PROJECT REPORT

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## DATASCRIBE AI

AI-Powered Unstructured Industrial & Inspection Data Structuring Platform

|               |                                                                                 |
|---------------|---------------------------------------------------------------------------------|
| Submitted By  | Phurushothaman M                                                                |
| Programme     | Bachelor of Technology (B.Tech) — Computer Science and Engineering (Final Year) |
| Institution   | Acharya College of Engineering Technology, Puducherry                           |
| Organisation  | Larsen & Toubro Limited — Construction                                          |
| Division      | Water & Effluent Treatment IC                                                   |
| Location      | TC-3, Tower-B, A-Wing, 5th Floor, Manapakkam, Chennai - 600089                  |
| Project Guide | Mr. Thiruvengadam — Project Manager, ISD                                        |
| Duration      | 19 March 2026 – 30 April 2026 (6 Weeks)                                         |
| Academic Year | 2025 – 2026                                                                     |

## 1. Internship Details

### 1.1 Host Organisation

Larsen & Toubro (L&T;) is one of India's largest conglomerates, founded in 1938, operating across Engineering, Construction, Manufacturing, Technology, and Financial Services in over 50 countries. L&T; Construction is India's largest construction company. The Water & Effluent Treatment IC manages critical water and sewerage infrastructure nationwide.

| Field         | Details                                                                     |
|---------------|-----------------------------------------------------------------------------|
| Company       | Larsen & Toubro Limited (L&T;)                                              |
| Business Unit | Construction — Water & Effluent Treatment IC                                |
| Address       | TC-3, Tower-B, A-Wing, 5th Floor, Manapakkam, Chennai – 600089              |
| Focus Area    | Water supply, sewerage, industrial effluent & drinking water infrastructure |

### 1.2 Supervision & Team

| Role          | Name & Designation                                                                   |
|---------------|--------------------------------------------------------------------------------------|
| ISD Head      | Mr. A. Srinivasan — Deputy General Manager, Information Systems Division             |
| Project Guide | Mr. Thiruvengadam — Project Manager, ISD                                             |
| Team Support  | Mr. K Siva Shunmugam (System Engineer), Ms. Kavitha (Asst. Manager)                  |
| Intern        | Phurushothaman M — B.Tech CSE Final Year, Achariya College of Engineering Technology |

### 1.3 Team Hierarchy

| Information Systems Division (ISD) — L&T; Construction               |
|----------------------------------------------------------------------|
| ▼                                                                    |
| Mr. A. Srinivasan   Deputy General Manager — Head, ISD               |
| ▼                                                                    |
| Mr. Thiruvengadam   Project Manager — ISD                            |
| ▼                                                                    |
| Mr. K Siva Shunmugam (System Engineer) & Ms. Kavitha (Asst. Manager) |
| ▼                                                                    |
| Phurushothaman M   Intern — B.Tech CSE Final Year                    |

## 2. Project Description

### 2.1 Abstract

DataScribe AI is a lightweight AI-powered web application developed to address a real operational challenge faced by engineers and field staff at L&T; Construction: converting unstructured industrial and inspection data into clean, structured, downloadable records.

Engineers typically record inspection findings, equipment status, and maintenance observations through voice dictation or free-form text in natural language. DataScribe AI processes this input — whether spoken or typed — and automatically constructs a formatted, organised table from it. Additionally, the system accepts pre-formatted blank inspection or maintenance forms and allows the engineer to simply dictate or type the field values in plain language; the AI understands the context, maps the answers to the correct fields, and fills the form automatically. Completed records can be downloaded as PDF or Excel files for reporting and archiving.

### 2.2 Modules

| Module                        | What It Does                                                                                                                                                                                                                                                               |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Structured Data Extraction | User dictates or types inspection/maintenance observations in natural language. The AI understands the content, identifies key data points, and constructs a clean, formatted table with appropriate columns and rows.                                                     |
| 2. Blank Form Auto-Fill       | User uploads a blank pre-formatted form (inspection checklist, maintenance sheet, etc.). The AI analyses the form fields, the user narrates or types the field values naturally, and the AI fills the blank spaces with the correct data — preserving the original layout. |
| 3. Export (PDF & Excel)       | The completed structured table or filled form can be downloaded instantly as a professional PDF report or an Excel spreadsheet for further analysis or submission.                                                                                                         |

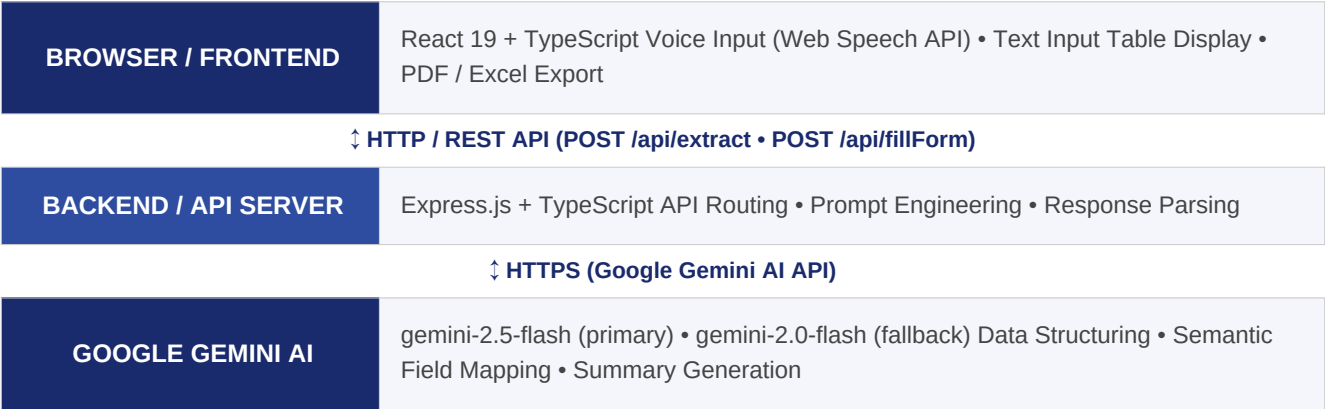
### 2.3 Problem Solved

| Before DataScribe AI                                       | After DataScribe AI                                          |
|------------------------------------------------------------|--------------------------------------------------------------|
| Engineer fills paper inspection forms manually.            | Engineer speaks or types; AI fills the form automatically.   |
| Data entry into Excel is slow and error-prone.             | Structured table generated instantly from natural language.  |
| Unstructured voice notes are not searchable or exportable. | Voice input converted to clean PDF/Excel in one click.       |
| Blank form fields left empty due to time pressure on site. | AI maps dictated values to correct form fields semantically. |

### 3. System Architecture & Data Flow

#### 3.1 High-Level Architecture

The system follows a clean three-layer architecture: a React frontend handling user input and export, an Express.js backend managing API routing and AI integration, and the Google Gemini AI service performing the data structuring and form mapping.



#### 3.2 Data Flow — Extraction Module

| Step | Action                                                     |
|------|------------------------------------------------------------|
| 1    | User speaks or types inspection/maintenance observations   |
| 2    | Browser captures input via Web Speech API or text field    |
| 3    | Frontend sends POST /api/extract to backend                |
| 4    | Backend builds a structured Gemini prompt with the input   |
| 5    | Gemini returns JSON: document type, columns, rows, summary |
| 6    | Backend sanitises and validates the JSON response          |
| 7    | Frontend renders the structured table and AI summary       |
| 8    | User downloads as PDF or Excel                             |

#### 3.3 Data Flow — Form Fill Module

| Step | Action                                                            |
|------|-------------------------------------------------------------------|
| 1    | User uploads blank inspection/maintenance form template           |
| 2    | System parses the form and identifies all blank fields            |
| 3    | User dictates or types field values in natural language           |
| 4    | Frontend sends POST /api/fillForm with template + narration       |
| 5    | Gemini semantically maps each narrated value to the correct field |
| 6    | Backend returns filled field mapping                              |
| 7    | Frontend renders the completed form preserving original layout    |

| Step | Action                                     |
|------|--------------------------------------------|
| 8    | User downloads filled form as PDF or Excel |

## 4. Technology Stack

| Layer        | Technology                       | Purpose                                              |
|--------------|----------------------------------|------------------------------------------------------|
| Frontend     | React 19 + TypeScript            | UI, state management, table rendering                |
| Build Tool   | Vite 6                           | Fast development server, optimised production builds |
| Styling      | Tailwind CSS 4                   | Utility-first responsive styling                     |
| Animations   | Motion (motion/react)            | Smooth UI transitions                                |
| Voice Input  | Web Speech API (native)          | Real-time browser speech-to-text, no SDK needed      |
| Backend      | Express.js 4 + TypeScript        | REST API server, AI integration, response parsing    |
| AI Service   | Google Gemini (gemini-2.5-flash) | Data structuring, semantic mapping, summaries        |
| PDF Export   | jsPDF + AutoTable (CDN)          | Professional PDF report generation in-browser        |
| Excel Export | SheetJS / xlsx (CDN)             | Excel file generation for downstream analysis        |
| Config       | Dotenv                           | Environment variable management                      |

## 5. Project Structure

|                |                                           |
|----------------|-------------------------------------------|
| datascribe-AI/ | — Project root                            |
| src/           | — Frontend source                         |
| App.tsx        | — Main React component (all UI + logic) ★ |
| main.tsx       | — React DOM entry point                   |
| index.css      | — Tailwind CSS imports                    |
| server.ts      | — Express backend + Gemini integration ★  |
| vite.config.ts | — Vite build configuration                |
| tsconfig.json  | — TypeScript compiler settings            |
| package.json   | — Dependencies & scripts                  |
| index.html     | — HTML entry point                        |
| .env           | — API keys & environment variables        |

## 6. Key Features

| Feature                 | Description                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------|
| Voice Dictation Input   | Engineer speaks inspection/maintenance data; browser captures it in real time.             |
| Text Input Mode         | Alternative to voice — user types observations in plain natural language.                  |
| AI Data Structuring     | Gemini AI interprets the narration and constructs a formatted table automatically.         |
| Blank Form Auto-Fill    | Upload a blank form; AI fills the empty fields from the user's narration.                  |
| Semantic Field Mapping  | AI understands meaning, not just keywords — handles synonyms and context naturally.        |
| Table-Aware Output      | Multi-column tabular forms and reports are handled with correct row/column mapping.        |
| AI Summary Generation   | A concise natural language summary of the extracted data is generated alongside the table. |
| PDF Export              | Completed table or filled form exported as a professional, print-ready PDF.                |
| Excel Export            | Data exported as .xlsx for further analysis, editing, or system upload.                    |
| AI Fallback Reliability | If the primary Gemini model fails, the system automatically retries with a fallback model. |



## 7. Installation & Setup

### 7.1 Prerequisites

| Tool           | Requirement   | Purpose                            |
|----------------|---------------|------------------------------------|
| Node.js        | v18+          | Backend Express server runtime     |
| npm            | Latest        | Package management                 |
| Browser        | Chrome / Edge | Web Speech API for voice input     |
| Gemini API Key | Required      | Google AI data structuring service |

### 7.2 Setup Steps

| Step | Action               | Command / Detail                                         |
|------|----------------------|----------------------------------------------------------|
| 1    | Clone repository     | git clone && cd datascribe-AI                            |
| 2    | Install dependencies | npm install                                              |
| 3    | Create .env file     | VITE_API_URL=http://localhost:3000 GOOGLE_GENAI_API_KEY= |
| 4    | Start dev server     | npm run dev → Frontend: :5173   Backend: :3000           |
| 5    | Build for production | npm run build → npm start                                |

## 8. System Evaluation

### 8.1 Advantages

| # | Advantage                      | Impact                                                            |
|---|--------------------------------|-------------------------------------------------------------------|
| 1 | Eliminates Manual Data Entry   | Field engineers save significant time on site paperwork.          |
| 2 | Natural Language Understanding | No training required — speak or type as you normally would.       |
| 3 | Semantic Form Filling          | Correct field mapping even when phrasing varies.                  |
| 4 | Dual Export Formats            | PDF for submission, Excel for analysis — one click each.          |
| 5 | AI Fallback Reliability        | Continuous availability even if the primary model is unavailable. |
| 6 | No Hardware Required           | Runs in any modern browser — no app installation needed.          |

### 8.2 Limitations & Mitigations

| Limitation          | Challenge                              | Mitigation                                     |
|---------------------|----------------------------------------|------------------------------------------------|
| Browser Dependency  | Web Speech API requires Chrome/Edge.   | Text input always available as fallback.       |
| No Data Persistence | Extracted data exists in session only. | Database integration planned for production.   |
| Internet Required   | Gemini API needs active connection.    | Dual model fallback; offline mode planned.     |
| No Authentication   | No user login or role control yet.     | JWT-based auth planned for production release. |

### 8.3 Evaluation Scorecard

| Dimension           | Rating   | Remarks                                                          |
|---------------------|----------|------------------------------------------------------------------|
| Extraction Accuracy | High     | Gemini handles varied industrial narration styles well.          |
| Form Fill Accuracy  | High     | Semantic mapping correctly fills most standard inspection forms. |
| Ease of Use         | High     | Voice-first; no technical expertise required from field staff.   |
| Export Quality      | High     | PDF and Excel outputs are clean and submission-ready.            |
| Scalability         | Moderate | Single Express server; containerisation needed for production.   |
| Offline Support     | Low      | AI features require internet; offline mode is a future goal.     |

## 9. Conclusion

DataScribe AI was developed during a six-week internship at Larsen & Toubro Construction — Water & Effluent IC, Chennai (19 March 2026 – 30 April 2026). The project addressed a real operational need: converting the unstructured voice and text observations recorded by field engineers into clean, structured, and downloadable inspection and maintenance records.

The system delivers two core capabilities — structured data extraction from natural language narration, and intelligent blank-form auto-filling — backed by Google Gemini AI and a full-stack TypeScript platform. Engineers can now speak or type their findings and receive a formatted table or completed form ready for PDF/Excel download in seconds.

| Module                     | Technology                        | Key Achievement                                                 |
|----------------------------|-----------------------------------|-----------------------------------------------------------------|
| Structured Data Extraction | React + Gemini AI                 | Converts voice/text narration into formatted tables instantly.  |
| Blank Form Auto-Fill       | Express + Gemini Semantic Mapping | Fills inspection/maintenance forms from natural language input. |
| Multi-Format Export        | jsPDF + SheetJS                   | PDF and Excel downloads, submission-ready in one click.         |

This mini-project demonstrates how a focused AI application can meaningfully reduce administrative burden for construction field teams, improve data quality, and accelerate reporting workflows — with a simple, browser-based interface that requires no technical expertise from end users.

## . Acknowledgements

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— With Gratitude, Respect & Warm Regards —

### **Mr. A. Srinivasan — Deputy General Manager, Head ISD**

“Your leadership created an environment where learning was purposeful and contribution was real. The direction you provided shaped every decision in this project. I am deeply honoured to have worked under your guidance.”

### **Mr. Thiruvengadam — Project Manager, ISD**

“Your mentorship went beyond technical instruction. You made complex problems approachable and every review session more valuable than a lecture. The professional standards you set are ones I will carry throughout my career.”

### **Mr. K Siva Shunmugam — System Engineer, ISD**

“Whenever I encountered a challenge, you responded with clear, patient, and thoughtful guidance. Your consistent support made this journey significantly more productive. Thank you sincerely.”

### **Ms. Kavitha — Assistant Manager, ISD**

“Your warmth and encouragement made the team feel like a place of genuine collaboration. This internship was enriched not just technically but personally because of you.”

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